ShopSmart Retail Group
Exploratory Data Analysis Report

Retail sales, customers, products, inventory and returns, 2025

Programme	AnalystLab Africa Experience Lab, Week 1
Mission	DS01: Exploratory Data Analysis on Retail Sales Data
Track	Data Science
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Client	ShopSmart Retail Ltd. (stores across Nigeria)
Date	October 2026

All revenue figures are in Nigerian Naira (NGN, assumed) and use Completed orders unless stated. The accompanying notebook (Week1_EDA.ipynb) reproduces every number in this report.

1. Executive Summary

ShopSmart asked for a clear picture of its retail data before any predictive models are built. This report analyses nine linked tables covering 50,000 orders placed in 2025 across 25 stores, 5,000 customers and 500 products. The data was cleaned, checked for internal consistency, explored statistically and visually, and used to engineer features for future machine learning.

Headline results. Realised revenue (Completed orders) is **NGN 2.50B** from 16,529 orders, with an average order value of NGN 151K. Only 33.1% of orders are Completed: a further NGN 2.55B sits in Cancelled and NGN 2.53B in Pending orders. Performance is strikingly even across stores, regions, categories and customer tiers, and statistical tests show the differences are consistent with random variation. The data also contains rule-breaking records (returns dated before the order, orders sold below cost, an ambiguous discount field) that should be fixed before any model is trained.

What we recommend. Fix the order-completion funnel, repair the data rules, restock 21 low-stock items while reviewing overstock, win back lapsed customers, and avoid reallocating resources on small differences until more history is available.

Key figures

Metric	Value
Total Revenue (Completed orders)	NGN 2.50B
Gross Order Value (all statuses)	NGN 7.58B
Total Orders (all statuses)	50,000
Completed Orders	16,529
Active Customers (>=1 completed order)	4,795
Average Order Value (AOV)	NGN 151K
Total Quantity Sold (units)	49,432
Return Rate	4.88%
Inventory Value (at cost)	NGN 3.17B

2. Dataset Description

The dataset has nine CSV tables and a data dictionary. Orders and Order_Items have a one-to-one relationship (each order contains one product), and all foreign keys resolve to a parent record.

Table	Rows	Columns	What it represents	Primary key
Customers	5,000	11	Registered customers: gender, age, city, state, registration date, membership level	customer_id
Products	500	7	Product catalogue: category, brand, cost price and selling price	product_id
Orders	50,000	12	Order headers: date, customer, product, store, quantity, price, discount, shipping, status, channel, payment	order_id
Order_Items	50,000	7	Line detail for each order, including line_total	order_item_id
Stores	25	6	25 stores: city, state, region, opening date	store_id
Returns	2,500	5	Returned orders: reason, date and refund amount	return_id
Employees	125	6	Store staff: role, hire date, salary	employee_id
Suppliers	489	6	Supplier names and contacts (linked to products through brand)	supplier_id
Inventory	500	6	Stock on hand, reorder level and last restock date per product and store	inventory_id

Note on the data dictionary. It holds 41 entries and covers only five tables (Orders, Customers, Products, Stores, Returns). Employees, Suppliers, Inventory and Order_Items have no entries, and the descriptions given are generic (for example, "Customer_ID field").

3. Data Cleaning Process

Every action was written to a cleaning log (Cleaned_Data/cleaning_log.csv). Where a problem could not be fixed with certainty, records were **flagged rather than deleted**, so that business totals are not silently changed.

Step	Finding	Action
Column names	Mixed upper and lower case	Converted to lower_snake_case; renamed ambiguous fields (membership_level, store_state)
Duplicates	No full-row or primary-key duplicates; 49 repeated customer emails; 58 orders with two return records	Nothing deleted; flagged
Missing values	None found, including hidden placeholders such as N/A or blank text	No imputation needed
Data types	All six date columns stored as text (M/D/YYYY)	Converted to datetime with zero failures; category dtype for low-cardinality fields
Inconsistent values	Categories are clean; phone numbers use many formats; 80 supplier phones too short	Digits-only column and validity flag added
Currency fields	All already numeric	No change
Integration	Nine tables joined on validated keys	Master dataset: 50,000 rows, row count asserted after every merge

Integrity and business-rule checks

Clean formatting is not the same as clean content. The following checks compare tables with each other and test business logic. The first block passed; the second shows failures that limit what the data can support.

Passed (14 checks): all 8 foreign-key checks, the agreement of Orders with Order_Items on product, quantity, price and discount, the line_total formula (quantity x price - discount) and catalogue prices always above cost.

Check that failed	Records	Share	Why it matters
order unit_price equals catalogue selling_price	50,000	100.0%	Order prices are unrelated to catalogue prices
order unit_price below product cost_price	12,163	24.3%	Would mean selling at a loss
return_date earlier than order_date	1,252	50.1%	A return before the purchase is impossible
return on a Cancelled or Pending order	1,685	67.4%	Only completed orders can be returned
refund_amount larger than line_total	270	10.8%	Refund exceeds what was paid
order placed before customer registration	5,159	10.3%	Customers appear to order before they exist; registration dates are unreliable

The discount field. Values (0, 5, 10, 15, 20) look like percentages, yet line_total is reproduced exactly by subtracting them as flat amounts. If they were meant as percentages, recorded revenue would be overstated by about NGN 250M (11%). ShopSmart should confirm which is correct. **Store identifiers.** Store names repeat (four "Lagos" stores) and regions do not follow geography, so store_id is used throughout.

4. Exploratory Findings

4.1 Business KPIs

KPI	Value	Definition
Total Revenue (Completed orders)	NGN 2.50B	Sum of line_total, Completed orders only
Gross Order Value (all statuses)	NGN 7.58B	Sum of line_total across all 50,000 orders
Total Orders (all statuses)	50,000	Rows in Orders
Completed Orders	16,529	order_status = Completed
Total Customers (registered)	5,000	Unique customer_id in Customers
Active Customers (>=1 completed order)	4,795	Unique customers in Completed orders
Total Products (catalogue)	500	Unique product_id in Products
Products Sold (Completed orders)	500	Unique products in Completed orders
Total Stores	25	Unique store_id in Stores
Average Order Value (AOV)	NGN 151K	Revenue / completed orders
Average Selling Price (per unit sold)	NGN 50K	Mean unit_price on Completed orders
Average Catalogue Selling Price	NGN 36K	Mean selling_price in Products
Total Quantity Sold (units)	49,432	Sum of quantity, Completed orders
Return Rate	4.88%	Orders with a return / all orders
Inventory Value (at cost)	NGN 3.17B	Sum of stock_on_hand x cost_price

4.2 Are the differences real?

Any ranking has a top and a bottom, even when gaps are random. Eight statistical tests (Kruskal-Wallis for order value, chi-square for returns) were run. **None found a significant difference** (all p-values above 0.05), so stores, categories, regions, membership levels and channels behave almost identically. The rankings below describe scale rather than quality.

Test	p-value	Conclusion
Kruskal-Wallis: order value by Store	0.282	Not significant
Kruskal-Wallis: order value by Category	0.721	Not significant
Kruskal-Wallis: order value by Membership level	0.706	Not significant
Kruskal-Wallis: order value by Region	0.141	Not significant
Kruskal-Wallis: order value by Sales channel	0.115	Not significant
Chi-square: return rate by Category	0.353	Not significant
Chi-square: return rate by Sales channel	0.252	Not significant
Chi-square: return rate by Store	0.242	Not significant

4.3 Descriptive statistics

Quantity is spread evenly from 1 to 5 and discount takes five fixed values, which is unusually uniform for real retail data. Order value is right-skewed (mean NGN 151K, median NGN 118K), so the median is the better measure of a typical order. About 0.6% of orders are statistical outliers; they are high-value orders, not errors, and are kept.

5. Visualisations

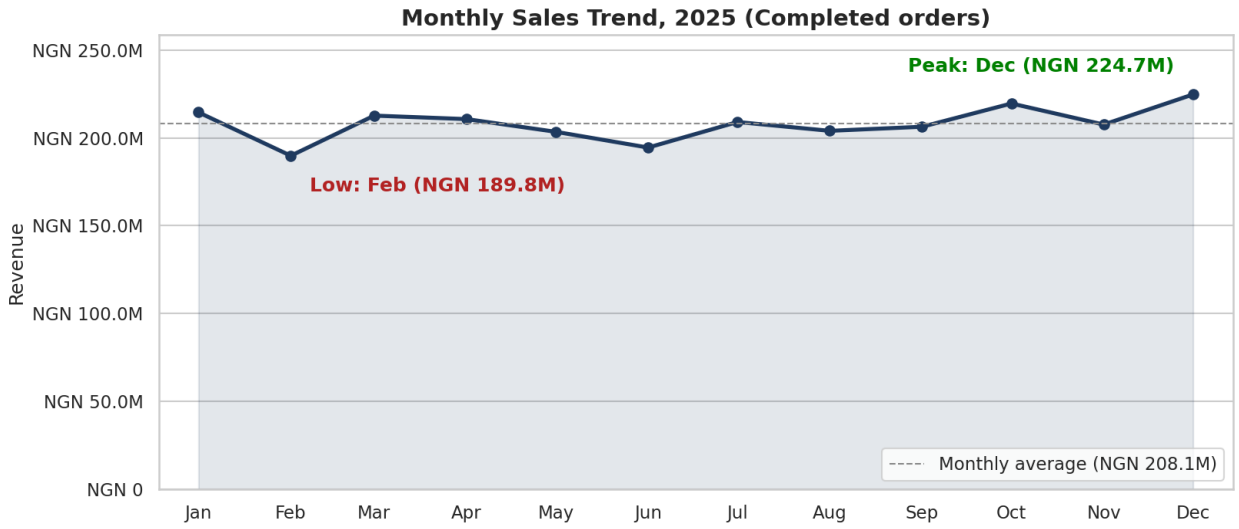


Figure 1. Revenue is steady near NGN 208M a month; December peaks and February is lowest (a 28-day month).

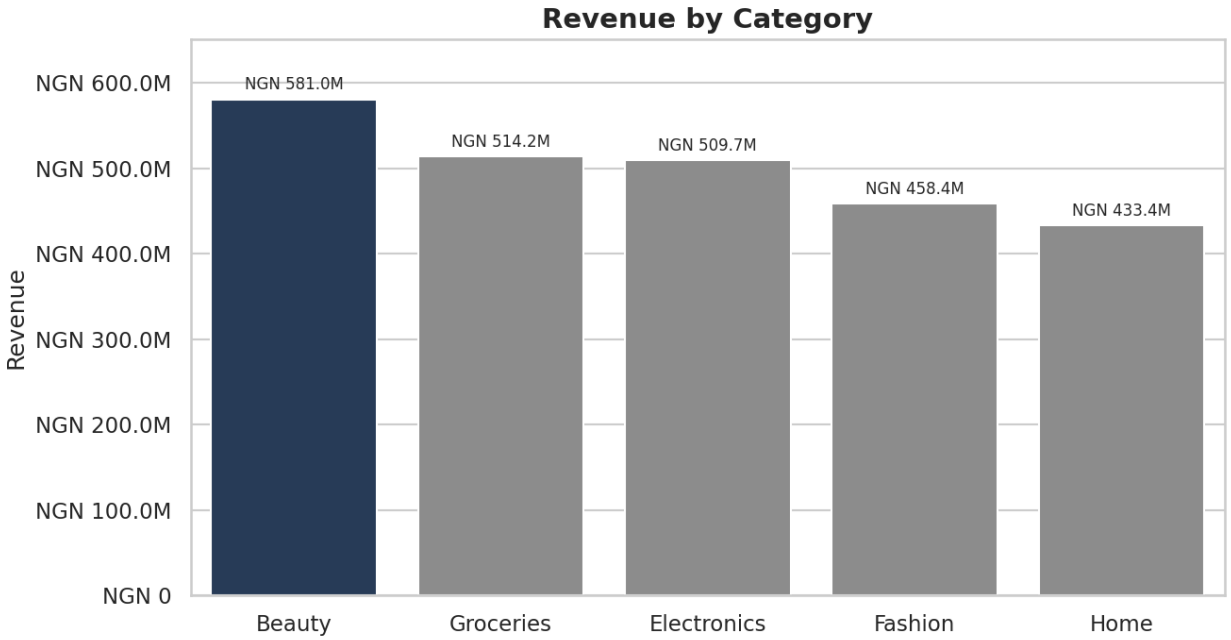


Figure 2. Beauty leads on revenue, mainly because it has the largest catalogue (119 products).

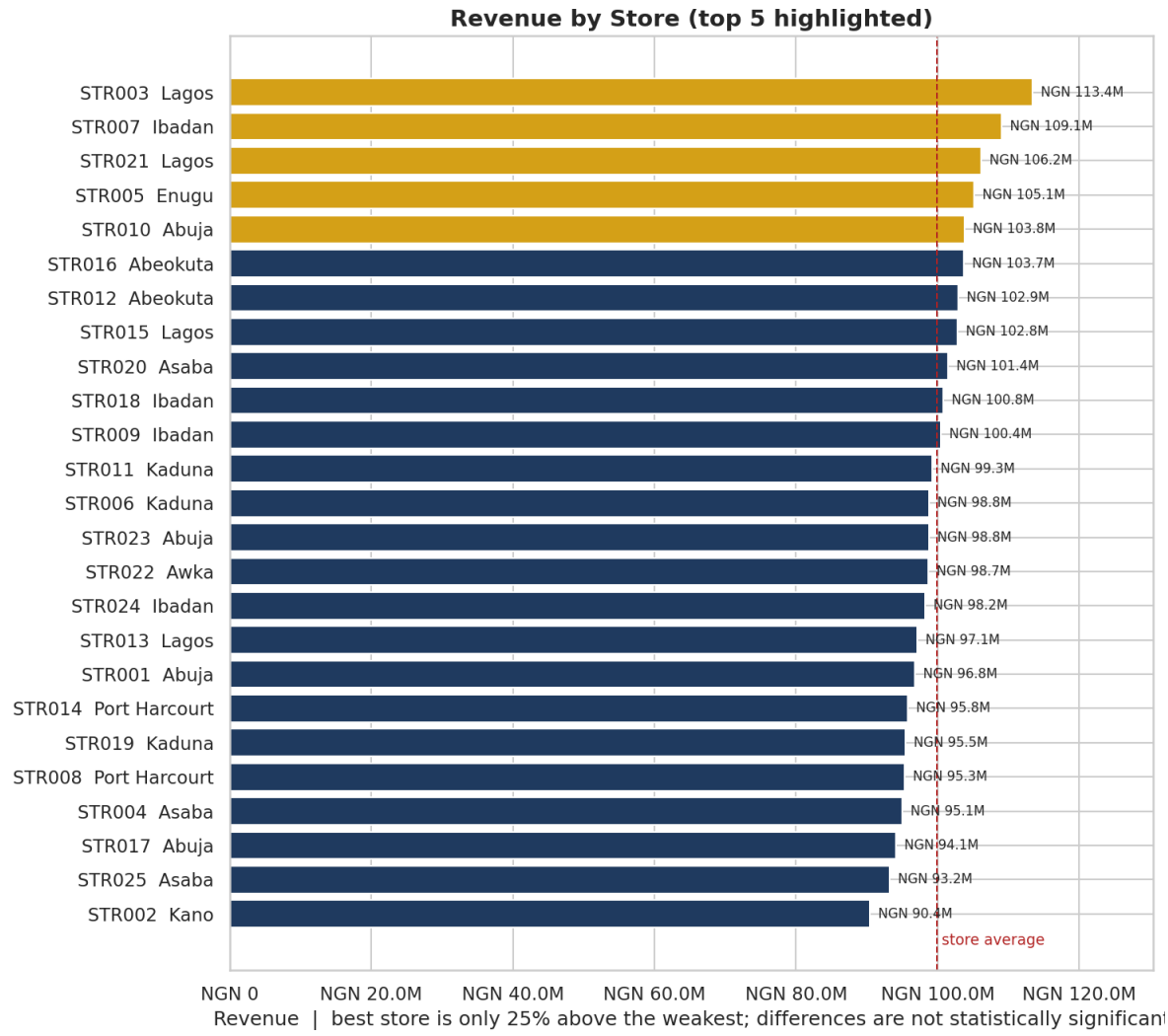


Figure 3. Revenue by store. The top store is 25% above the lowest, but the gaps are within random variation.

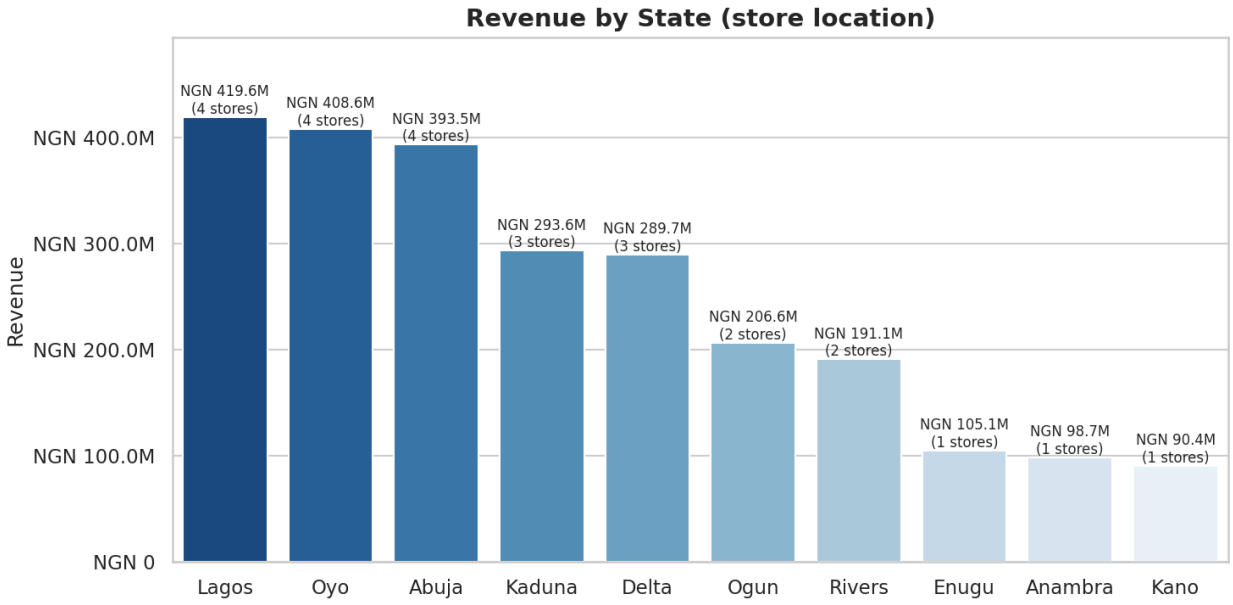


Figure 4. State rankings follow store counts; revenue per store is nearly flat (about NGN 90M to 105M).

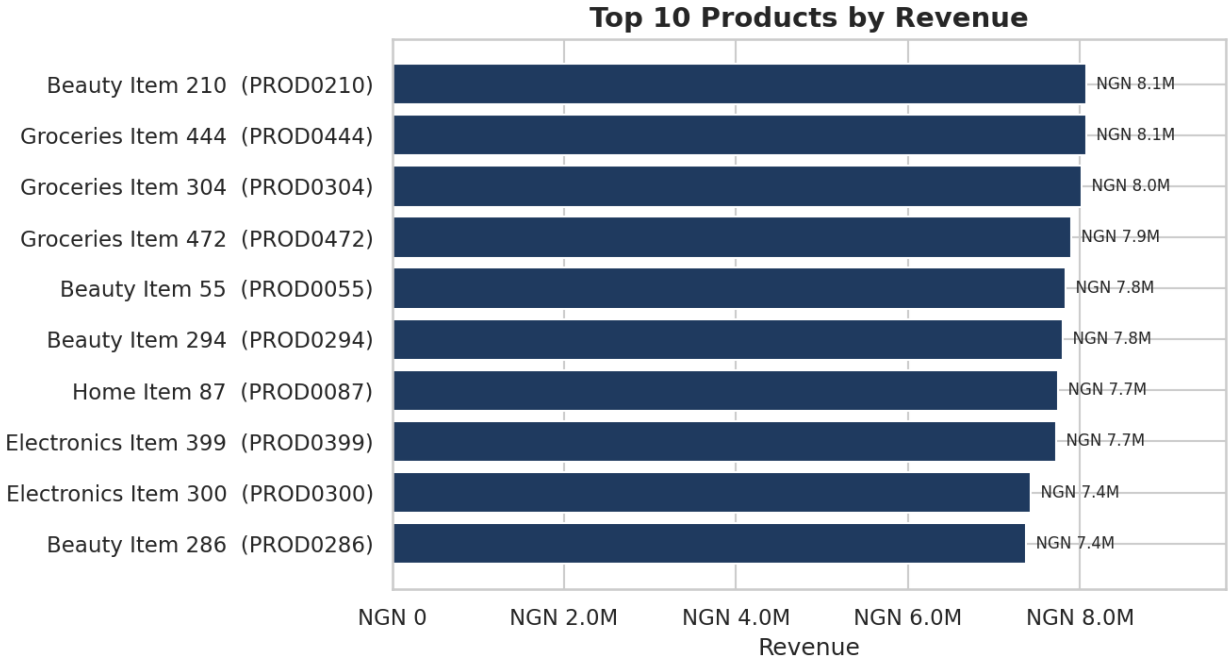


Figure 5. Top 10 products. No product exceeds 0.32% of revenue; ShopSmart does not rely on hero products.

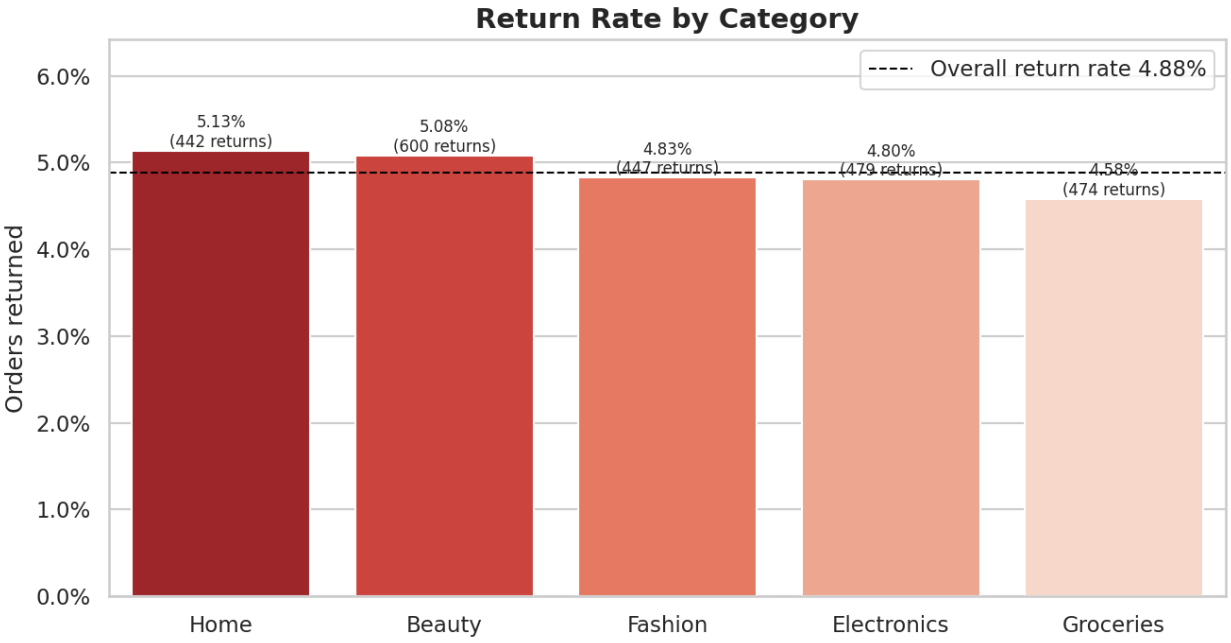


Figure 6. Return rates are 4.6% to 5.1% in every category, close to the 4.88% overall rate.

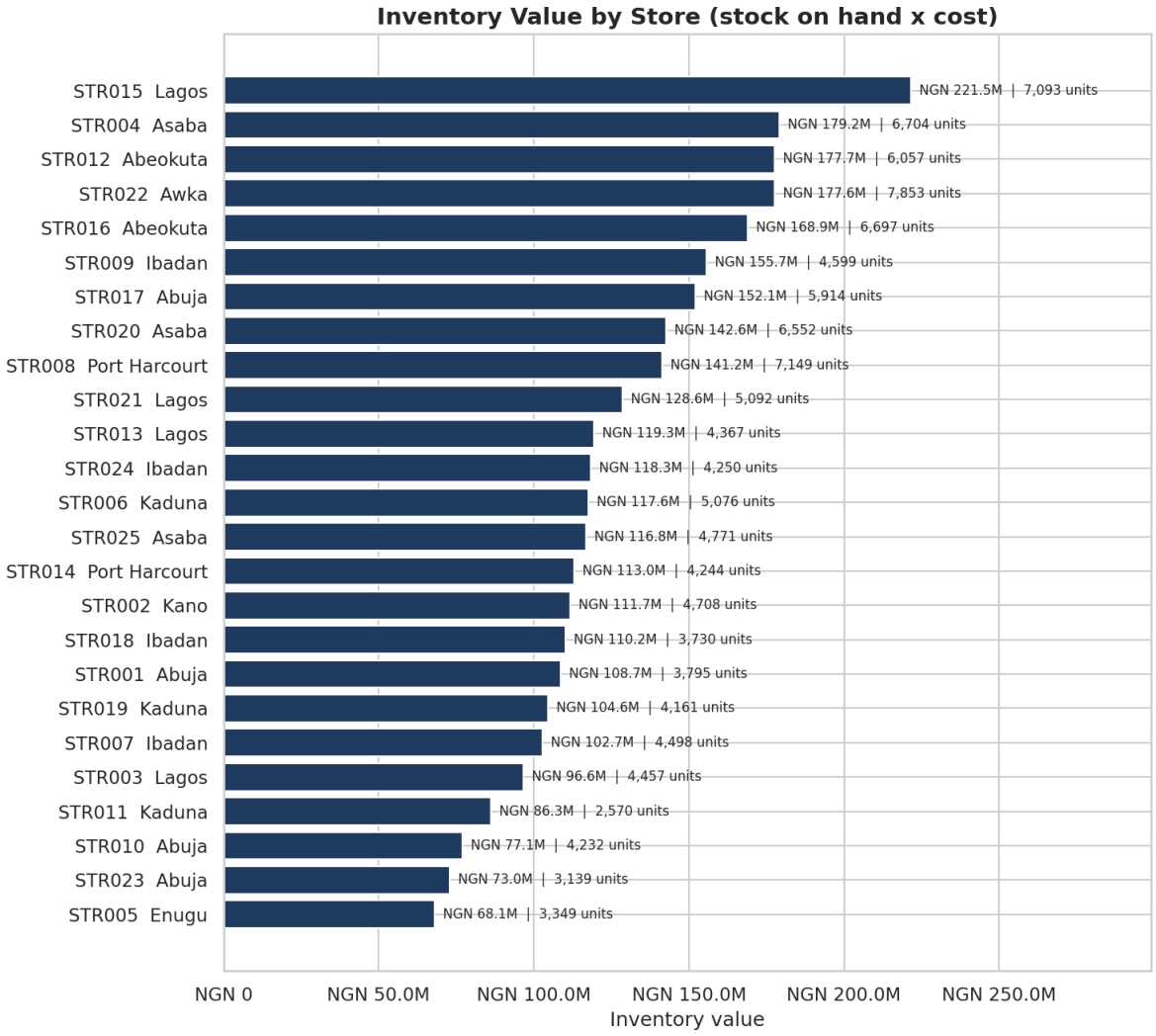


Figure 7. Inventory value ranges from NGN 68M to NGN 221M per store.

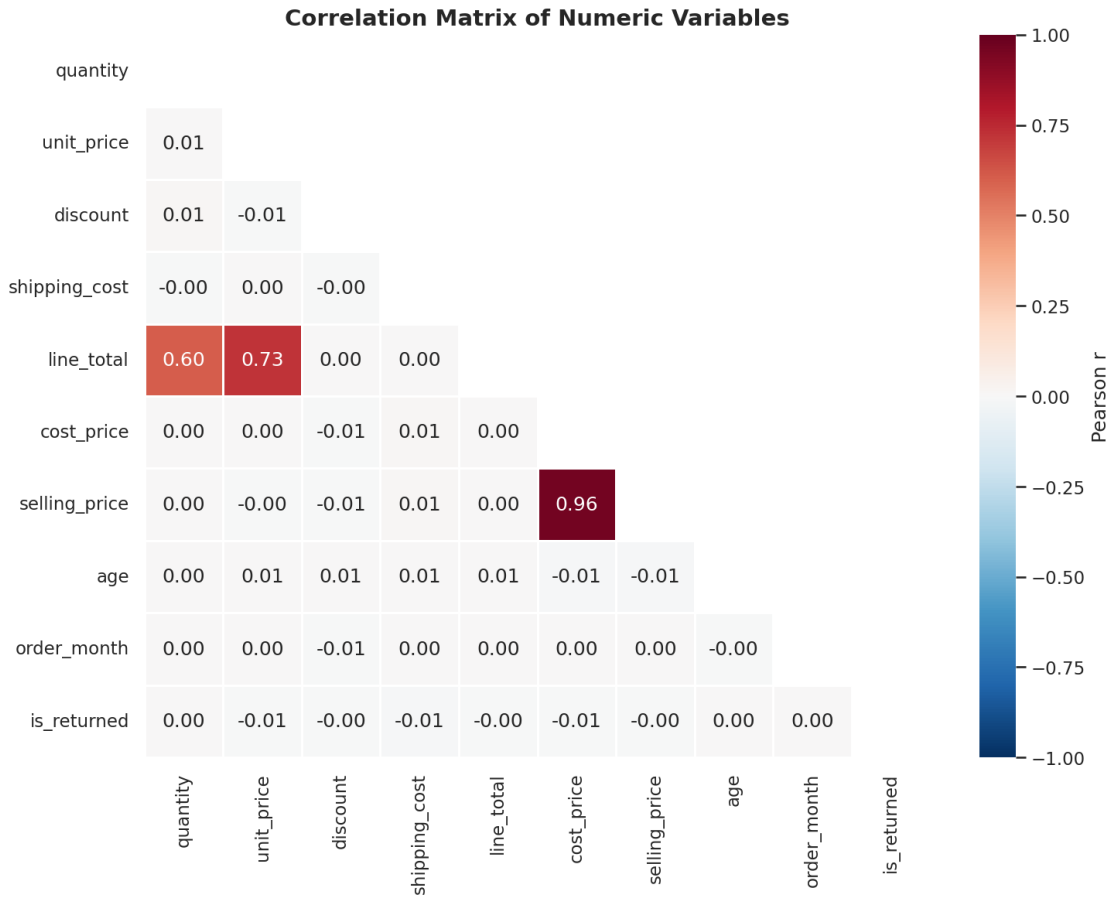


Figure 8. Only mechanical relationships appear (price vs cost, line total vs price and quantity); all else is near zero.

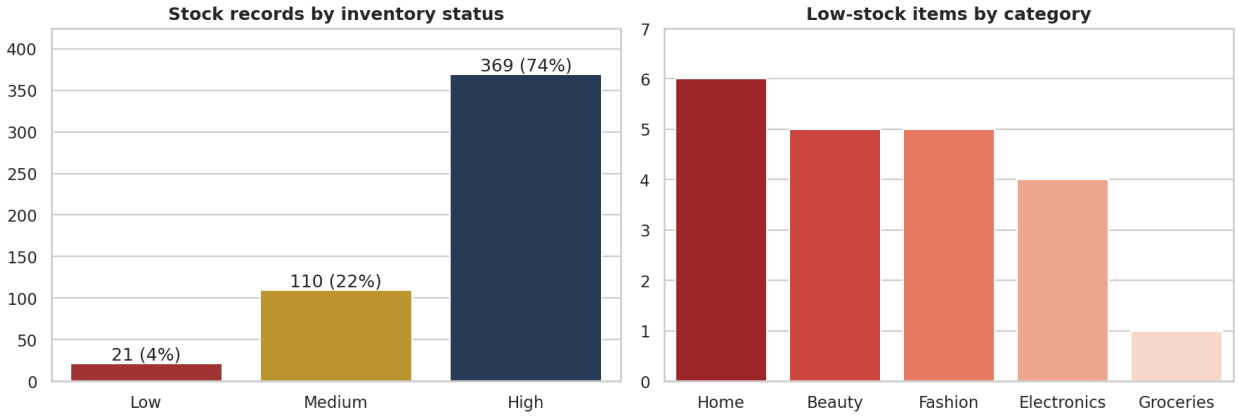


Figure 9. 21 stock records are at or below reorder level, while 74% hold more than five times their reorder level.

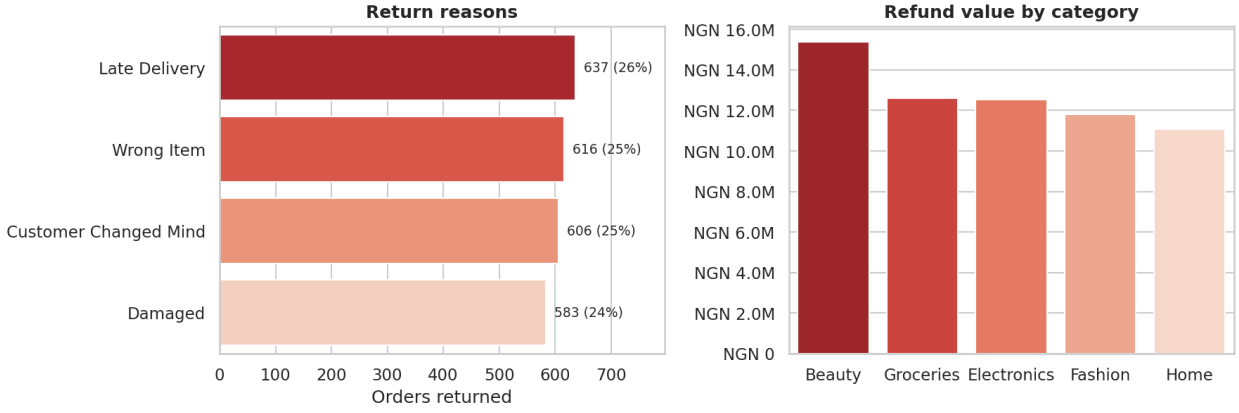


Figure 10. Return reasons are evenly split; about 75% are operational causes ShopSmart can influence.

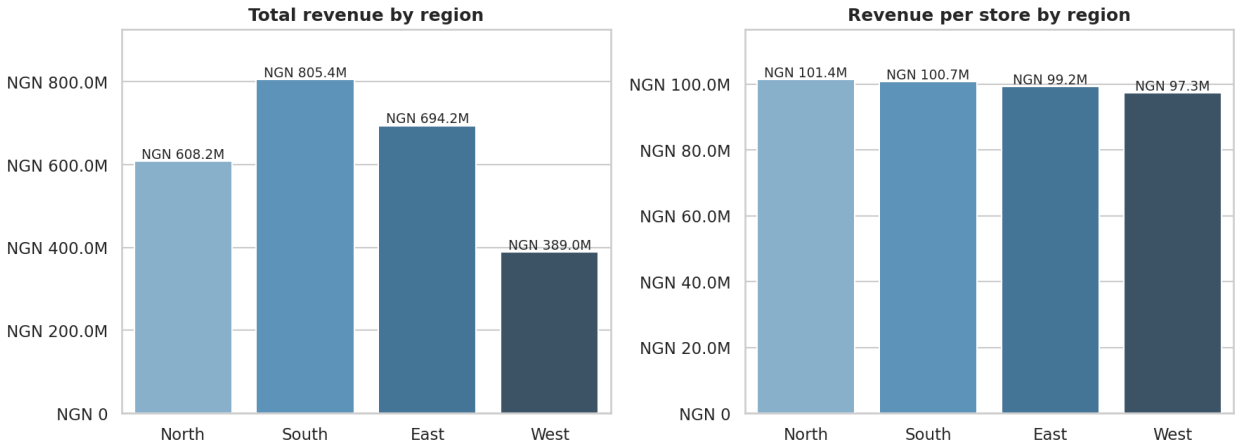


Figure 11. The West is lowest in total only because it has 4 stores; per store the regions are within 4.2%.

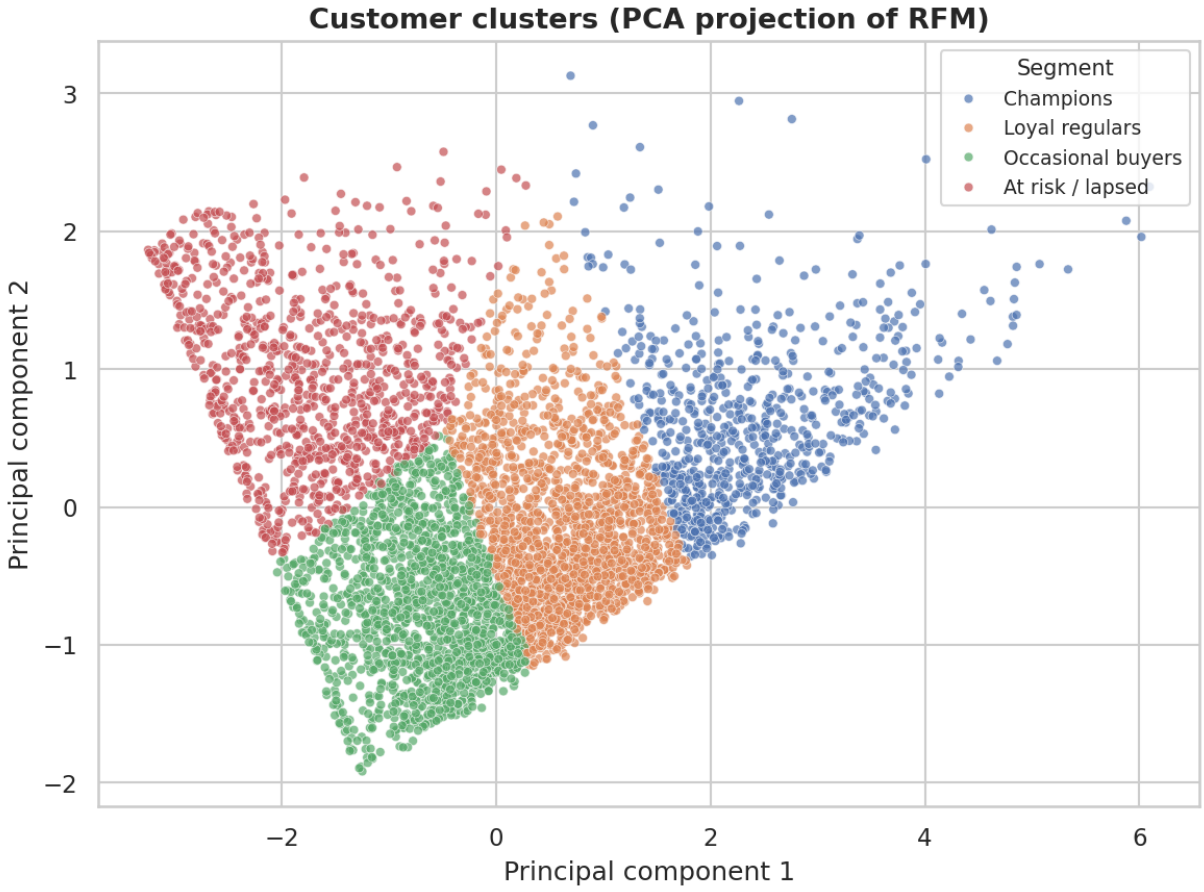


Figure 12. K-Means customer segments on recency, frequency and spend (PCA projection).

6. Answers to the Ten Business Questions

Question	Answer
1. Products with most revenue	Beauty Item 210 (NGN 8.1M), then Groceries Item 444 and 304. Top 10 are only 3.1% of revenue. Beauty leads as a category through catalogue size; Electronics earns the most per product.
2. Best stores	STR003 Lagos (NGN 113.4M, +13.6% vs average); weakest STR002 Kano (NGN 90.4M). Order counts do not differ beyond chance (p = 0.53).
3. Customer segments that spend most	Segments are close: tiers within 0.6%, gender 4.3%, age 4.6%. State gap is widest (Ogun vs Delta, 15.3%). Top 10% of customers give 24.8% of revenue.
4. Products with high return rates	No product stands out: top rates (about 11%) come from about 10 returns each and are consistent with chance (p = 0.95). 75% of returns have operational causes.
5. Peak sales months	December (NGN 224.7M), October, January. February is lowest but is a short month. Q4 is the strongest quarter; Thursday the busiest day.
6. Growing categories	Fashion (+10.3% Q4 vs Q1), Beauty (+8.1%), Home, Electronics, Groceries (+1.3%). No trend is statistically significant.
7. Best suppliers	478 of 489 suppliers supply one product, so none stands out on volume. Top by revenue: Thomas Ltd Supplies (USA, NGN 11.5M). No delivery data exists.
8. Items to restock	21 records at or below reorder level, led by Home Items 212, 414, 302 and Electronics Item 165. Overstock (74% of records) is the bigger problem.
9. Underperforming regions	None materially. Per store the regions are within 4.2% (North NGN 101.4M to West NGN 97.3M).
10. Patterns for predictive modelling	Order value is arithmetic (R-squared 1.00). Returns are unpredictable with current data (AUC 0.49). Customer behaviour (RFM) shows real structure.

7. Feature Engineering

Features were built at order, customer and inventory level. They are saved in Cleaned_Data and wrapped in re-usable functions.

Feature	Why it may improve future models
Profit margin (catalogue) and transaction margin	Separates profitable from low-value sales; flags sales below cost
Order month, quarter, weekday, weekend flag	Lets forecasting models learn calendar patterns
Customer age group	Stable segments instead of exact age
Discount percentage	Puts discounts on one scale across cheap and expensive items
Customer lifetime value (approximate)	Average order value x purchases per year x 3-year lifespan; ranks long-run customer worth
Days since last purchase (recency), frequency, spend	Classic churn and segmentation inputs (RFM)
Inventory status (Low / Medium / High), stock cover ratio	Ready-made label and urgency score for restocking models

Customer segmentation (bonus)

K-Means on recency, frequency and spend produced four segments. The silhouette score of 0.32 means the groups are slices of a continuum rather than natural clusters; four were chosen because they are more useful to the business than two (silhouette 0.37).

Segment	Customers	Orders each	Avg spend	Days since last buy	Revenue share
Champions	691 (14.4%)	6.4	NGN 1.1M	50	31.0%
Loyal regulars	1,652 (34.5%)	4.1	NGN 633K	62	41.9%
Occasional buyers	1,498 (31.2%)	2.3	NGN 277K	67	16.6%
At risk / lapsed	954 (19.9%)	2.0	NGN 275K	233	10.5%

8. Five Key Insights

1. Only one order in three is completed. 33.1% of orders are Completed, 33.7% Cancelled and 33.2% Pending. Cancelled orders hold NGN 2.55B and Pending orders NGN 2.53B of order value, against NGN 2.50B realised. Each additional percentage point of completion is worth about NGN 76M.

2. Performance is remarkably even. No statistical test found a real difference between stores, categories, regions, tiers or channels. The best store is 13.6% above average, regions are within 4.2% per store, and revenue per product by category ranges only from NGN 4.88M to NGN 5.15M.

3. Data reliability limits what can be concluded. The discount field may overstate revenue by about NGN 250M (11%); 24.3% of orders are priced below product cost; about half of returns pre-date their order; 67% of returns belong to non-completed orders; and 270 refunds exceed the order value.

4. Customer value is moderately concentrated and loyalty tiers do not work. The top 10% of customers generate 24.8% of revenue, and Gold, Silver and Bronze customers spend within 0.6% of each other. 14.4% of customers (Champions) deliver 31.0% of revenue, while 19.9% have been inactive for about 233 days.

5. Inventory is out of balance. 21 records (4.2%) are at or below reorder level, yet 73.8% hold more than five times their reorder level, representing NGN 2.95B of the NGN 3.17B inventory value.

Other observations: Q4 is the strongest quarter (about 6% above the Q1 to Q3 average), December the peak month and Thursday the busiest day. About 75% of returns come from late delivery, wrong item or damage.

9. Five Recommendations

1. Fix the order-completion funnel first. Record a reason for every cancellation, set a time limit for Pending orders and track the completion rate weekly. This is the largest revenue opportunity in the data.

2. Fix the data rules before building models. Confirm how discounts work and where order prices come from, validate returns (after the order, Completed orders only, refund no larger than the sale), give stores unique names and real regions, add supplier_id to Products and complete the data dictionary.

3. Restock the 21 low-stock items now and review overstock. Start with Home Items 212, 414 and 302 and Electronics Item 165, and set an automatic alert based on inventory status. Review high-cover stock to release working capital.

4. Win back lapsed customers and redesign the membership tiers. Target the roughly 950 At-risk customers with a re-engagement offer, protect the Champions, and make Gold, Silver and Bronze reflect real spending.

5. Plan around Q4 and December, but do not reallocate resources on small differences. Prepare for the seasonal lift and weekday pattern, but wait for multi-year data before moving stock or budget between stores, regions or categories.

10. Suggested Machine Learning Problems

Problem	Feasible now?	Notes
Customer segmentation	Yes	RFM shows real structure; demonstrated with K-Means.
Customer churn prediction	Partly	Features ready, but needs a churn definition (for example no purchase in 180 days) and more than one year of history.
Sales forecasting	Partly	Works at aggregate level; one year cannot confirm seasonality. Two to three years are needed.
Inventory forecasting	Not yet	Only one stock snapshot; no stock-movement history.

Problem	Feasible now?	Notes
Product recommendation	Limited	One item per order means no basket data; customers average 3.3 completed orders.
Return prediction	No	ROC-AUC is 0.49 (a coin flip) and the returns data has logic errors. Fix the data first.
Anomaly detection	Yes (rules first)	Below-cost sales, refunds above the sale and returns before purchase can be caught with simple rules.

11. Limitations and Assumptions

- One year of data (2025) that appears synthetic or heavily simplified (uniform quantities, discounts, channels and statuses), so patterns may be weaker than in a real retail business.
- Currency is assumed to be NGN. Revenue counts Completed orders only; the gross alternative is NGN 7.58B.
- No delivery, lead-time, promotion or basket-level fields exist, which limits supplier, return and recommendation analysis.
- Return analysis carries the data-quality caveats described in Section 3.
- Rankings of stores, categories and regions are reported but should not be read as evidence of real performance differences.